

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 28 — Chemistry of Transition Elements

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Electronic Configurations · Complex Ions & Ligand Exchange · Stereoisomerism · d-Orbital Splitting & Colour
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

- 28.1 Transition Element Characteristics & Redox Chemistry** — Definition of a transition element; anomalous electronic configurations of Cr and Cu; variable oxidation states
- 28.2 Complex Ions, Ligand Substitution & Chelate Effect** — Monodentate, bidentate (1,2-diaminoethane, ethanedioate), and polydentate (EDTA⁴⁻) ligands; chelate effect
- 28.3 Stereoisomerism in Transition Metal Complexes** — Geometric (cis-trans) isomerism in square planar complexes (cisplatin [Pt(NH₃)₂Cl₂] and its anti-isomer) and octahedral complexes
- 28.4 d-Orbital Splitting, Light Absorption & Colour** — Crystal field theory; splitting of degenerate 3d orbitals in octahedral (t_{2g} and e_g) and tetrahedral complexes
- 28.5 Homogeneous & Heterogeneous Catalysis** — Role of transition metals and ions as catalysts; variable oxidation states in homogeneous catalysis
- High-Frequency Core Repeats (Q41–Q50)** — The 10 most frequently tested Cambridge Paper 4 questions on Transition Elements from the past 10 years

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 d-Orbital Splitting & Origin of Colour in Octahedral Complexes — 9701/42/M/J/23/Q5 [6 Marks]

HARD · Syllabus 28.4

Aqueous copper(II) ions exist as hexaaquacopper(II), $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, which appears pale blue. The splitting of the 3d orbitals in an octahedral complex is shown in Fig. 1.1. Data: Planck constant $h = 6.63 \times 10^{-34} \text{ J s}$; Speed of light $c = 3.00 \times 10^8 \text{ m s}^{-1}$.

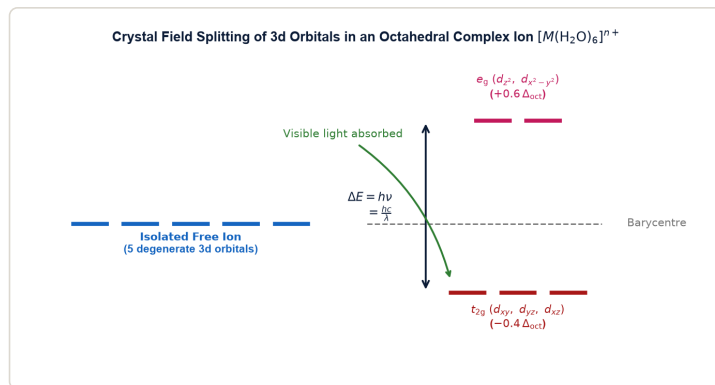


Fig. 1.1: Octahedral crystal field splitting of 3d orbitals showing electron promotion.

- (a) Explain why isolated gaseous Cu^{2+} ions have five degenerate 3d orbitals, but in $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ the 3d orbitals split into two energy levels. [2]

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- (b) Explain in detail why $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ appears pale blue in white light. [2]

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- (c) The wavelength of maximum absorption (λ_{max}) for $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ is 650 nm. Calculate the orbital splitting energy, E , in kJ mol^{-1} . [2]

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2 Cisplatin vs Transplatin Stereoisomerism & DNA Binding — 9701/41/M/J/23/Q5 [6 Marks]

HARD · Syllabus 28.3

The platinum complex $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ exists as two square planar geometric isomers, as illustrated in Fig. 2.1. Cisplatin is widely prescribed as a chemotherapy medication for treating testicular and ovarian cancers.

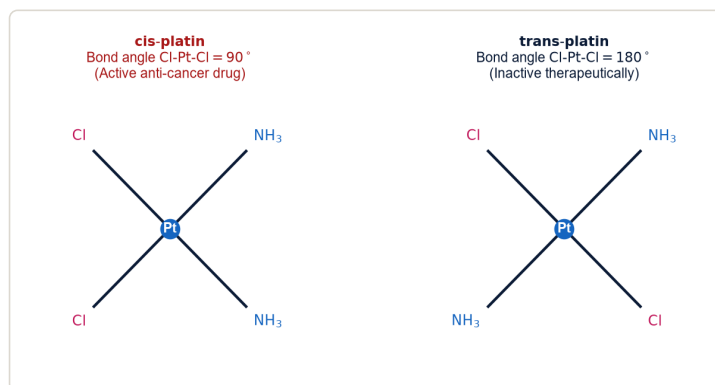


Fig. 2.1: Structures of square planar cisplatin and transplatin stereoisomers.

- (a) Draw 3D diagrams of cisplatin and transplatin, showing bond angles and dipole moments. [2]
- (b) Explain how cisplatin acts as an anti-cancer drug by binding to cellular DNA. [2]
- (c) Explain why transplatin cannot bind to DNA in the same manner and is completely ineffective as an anti-cancer drug. [2]

3 Visible Light Absorption Spectrum & Ligand Field Strength — 9701/42/O/N/23/Q5 [6 Marks]

HARD · Syllabus 28.4

The visible absorption spectrum of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ is shown in Fig. 3.1.
When excess concentrated ammonia is added, the solution turns deep royal blue as $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ forms, and λ_{max} shifts to 600 nm.

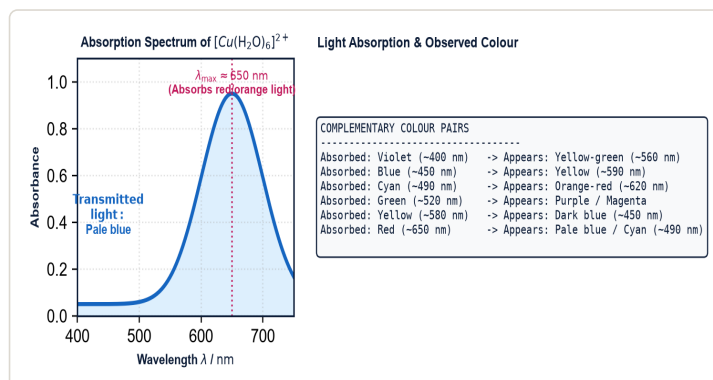


Fig. 3.1: Visible absorption spectrum of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ and complementary colour relationship.

- (a) Explain why the addition of ammonia causes a shift in the absorption maximum towards a shorter wavelength (higher frequency). [2]
- (b) Calculate the energy gap E in J per ion for $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ ($\lambda_{\text{max}} = 600 \text{ nm}$). [2]
- (c) Explain why transition metal complexes containing Sc^{3+} or Zn^{2+} ions are completely colourless in aqueous solution. [2]

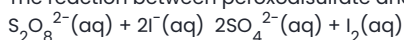
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- ## 5 Optical Isomerism in Octahedral Cobalt Complexes — 9701/42/M/J/22/Q5 [6 Marks] HARD · Syllabus 28.3

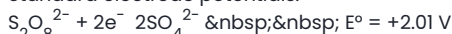
The complex ion $[\text{Co}(\text{en})_3]^{3+}$ contains cobalt(III) coordinated to three bidentate 1,2-diaminoethane ligands ('en' = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$). It exhibits optical isomerism.

- 6 Homogeneous Catalysis Mechanism: $\text{Fe}^{3+}/\text{Fe}^{2+}$ and Persulfate-Iodide — 9701/41/M/J/22/Q5 [6 Marks] HARD · Syllabus 28.5

The reaction between peroxodisulfate and iodide is thermodynamically feasible but kinetically very slow:



Standard electrode potentials:



- MENTORA ACADEMY · Urwah · A Level Theory Pack (Paper 4) – Topic 28 – Chemistry of Transition Elements

- (b) Write two balanced chemical equations showing how Fe^{3+} ions catalyse this reaction. [2]

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- (c) Calculate E°_{cell} for each of the two catalytic steps to demonstrate that both are energetically feasible. [2]

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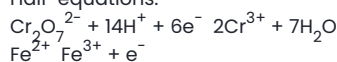
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7 Redox Titration: Iron(II) in Lawn Sand with Acidified Dichromate — 9701/42/O/N/22/Q5 [6 Marks]

HARD · Syllabus 28.1

A 2.50 g sample of lawn sand containing iron(II) sulfate was dissolved in dilute sulfuric acid to make 250 cm^3 of solution. A 25.0 cm^3 aliquot required 22.4 cm^3 of $0.0200 \text{ mol dm}^{-3} \text{ K}_2\text{Cr}_2\text{O}_7$ for complete oxidation using a barium diphenylaminesulfonate redox indicator.

Half-equations:



- (a) Construct the overall balanced ionic equation for the titration reaction. [2]

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- (b) Calculate the total mass of FeSO_4 ($M_r = 151.9$) in the 2.50 g lawn sand sample. [3]

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- (c) Calculate the percentage by mass of FeSO_4 in the lawn sand. [1]

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8 Geometry and Coordination Number of Cobalt Complexes — 9701/41/O/N/22/Q5 [6 Marks]

HARD · Syllabus 28.2

Cobalt forms several coordination complexes with different geometries:

- Complex A: $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (pink)
- Complex B: $[\text{CoCl}_4]^{2-}$ (blue)
- Complex C: $[\text{Co}(\text{NH}_3)_6]^{3+}$ (yellow-brown)

- (a) State the coordination number, geometry, and bond angles for Complex A and Complex B. [2]

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- (b) Explain why Complex A is octahedral while Complex B is tetrahedral, considering the properties of the ligands. [2]

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- (c) State the electronic configuration of cobalt in Complex A and in Complex C. [2]

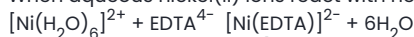
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9 Thermodynamics of the Chelate Effect: EDTA⁴⁻ Complexation — 9701/42/M/J/21/Q5 [6 Marks]

HARD · Syllabus 28.2

When aqueous nickel(II) ions react with hexadentate EDTA⁴⁻, all six water ligands are displaced:



At 298 K, $H^\circ = -34.0 \text{ kJ mol}^{-1}$ and $S^\circ = +188 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Explain what is meant by a hexadentate ligand. [2]

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- (b) Explain why the entropy change, S° , for this reaction is large and positive. [2]

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- (c) Calculate G° for the reaction at 298 K and explain why $[\text{Ni}(\text{EDTA})]^{2-}$ has a remarkably high stability constant ($K_{\text{stab}} \approx 10^{18} \text{ dm}^3 \text{ mol}^{-1}$). [2]

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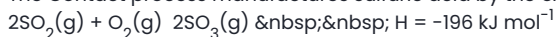
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10 Heterogeneous Catalysis: Vanadium(V) Oxide in the Contact Process — 9701/41/M/J/21/Q5 [6 Marks]

HARD · Syllabus 28.5

The Contact process manufactures sulfuric acid by the oxidation of sulfur dioxide:



Solid vanadium(V) oxide, V_2O_5 , acts as the catalyst at 450 °C.

- (a) Describe the stages of heterogeneous catalysis: adsorption, bond weakening, reaction, and desorption. [3]

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- (b) Write two two-step chemical equations demonstrating how vanadium changes oxidation state during the catalytic cycle. [2]

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- (c) Explain why 450 °C is chosen as an operational compromise temperature. [1]

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11 Electronic Configurations of Transition Atoms and Ions — 9701/42/O/N/21/Q5 [6 Marks]

HARD · Syllabus 28.1

The transition metals Scandium to Zinc occupy Period 4 of the Periodic Table.

- (a) Write the full electronic configurations of a chromium atom (Cr) and a copper atom (Cu), and explain why they are anomalous. [2]
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- (b) Write the electronic configuration of a Fe^{2+} ion and a Fe^{3+} ion, stating which is more stable and why. [2]
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- (c) Explain why zinc is not classified as a transition element, whereas copper is. [2]
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12 Titration of Ethanedioate with Manganate(VII) & Autocatalysis — 9701/41/O/N/21/Q5 [6 Marks]

HARD · Syllabus 28.5

A 25.0 cm^3 sample of sodium ethanedioate, $\text{Na}_2\text{C}_2\text{O}_4$, was acidified with dilute sulfuric acid and titrated against $0.0200 \text{ mol dm}^{-3}$ KMnO_4 at 60°C .

Reaction: $2\text{MnO}_4^- + 16\text{H}^+ + 5\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 10\text{CO}_2$

Titre volume = 24.5 cm^3 .

- (a) Calculate the concentration of the sodium ethanedioate solution in mol dm^{-3} . [3]
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- (b) Explain why no indicator is required in this titration. [1]
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- (c) Explain why the reaction starts very slowly before accelerating rapidly, identifying the autocatalytic species. [2]
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13 Chromium Chemistry: Amphoteric Hydroxide & Oxidation of Cr(III) to Cr(VI) — 9701/42/M/J/20/Q5 [6 Marks]

HARD · Syllabus 28.1

Chromium displays a wide variety of oxidation states and amphoteric behavior.

- (a) Describe what is observed when aqueous sodium hydroxide is added dropwise until in excess to aqueous chromium(III) sulfate, writing equations for both reactions. [3]
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- (c) Explain how the yellow chromate(VI) solution can be converted into an orange dichromate(VI) solution. [1]

HARD · Syllabus 28.2

$$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) + \text{SCN}^{-}(\text{aq}) \rightleftharpoons [\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$$

(a) Write the expression for the equilibrium constant K_c (or K_{stab}) for this reaction. [2]

- (c) State the type of reaction occurring, and explain why the coordination number remains 6. [2]

HARD · Syllabus 28.2

- Complex 1: Tetracarbonylnickel(0), $\text{Ni}(\text{CO})_4$ (tetrahedral, volatile)

- Complex 2: Tetracyanonickelate(II), $[\text{Ni}(\text{CN})_4]^{2-}$ (square planar)

- (c) $\text{Ni}(\text{CO})_4$ is used in the Mond process for refining nickel. Explain how this process purifies nickel metal. [2]

16 d-d Transitions vs Charge Transfer: Why Permanganate is Deep Purple — 9701/41/O/N/19/Q5 [6 Marks] **HARD · Syllabus 28.4**

Transition metal colours arise from electronic transitions.
Manganate(VII), MnO_4^- , possesses an intensely vibrant deep purple colour.

- (a) Determine the oxidation state of manganese in MnO_4^- and write its electronic configuration. **[2]**

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- (b) Explain why the purple colour of MnO_4^- cannot arise from d-d transitions. **[2]**

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- (c) State the origin of the intense colour in MnO_4^- . **[2]**

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Definition of a Transition Element & Non-Transition Elements — 9701/42/M/J/23/Q6 [4 Marks] **MEDIUM · Syllabus 28.1**

The d-block contains elements with filling d-orbitals.

- (a) Define the term transition element. **[2]**

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- (b) Explain why scandium (Sc) is not classified as a transition element. **[2]**

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18 Definition of Ligand and Coordination Number — 9701/41/M/J/23/Q6 [4 Marks] **MEDIUM · Syllabus 28.2**

Transition metal complexes contain ligands surrounding a central metal ion.

- (a) Define the term ligand. **[2]**

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- (b) Define the term coordination number. **[2]**

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19 Bidentate Ligands: Ethanedioate Complexation — 9701/42/O/N/23/Q6 [4 Marks] **MEDIUM · Syllabus 28.2**

Ethanedioate, $\text{C}_2\text{O}_4^{2-}$, acts as a bidentate ligand.

- (a) Draw the structural formula of the ethanedioate ion, indicating the two coordinating oxygen atoms with lone pairs. **[2]**

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- (b) State the formula, charge, and coordination number of the complex formed between Fe^{3+} and three ethanedioate ligands. [2]
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20 Tetrahedral vs Square Planar Complexes of Nickel(II) — 9701/41/O/N/23/Q6 [4 Marks]

MEDIUM · Syllabus 28.2

Nickel(II) forms $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$.

- (a) State the geometry and bond angles of $[\text{NiCl}_4]^{2-}$. [2]
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- (b) State the geometry and bond angles of $[\text{Ni}(\text{CN})_4]^{2-}$. [2]
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21 Optical Isomerism in Octahedral $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ — 9701/42/M/J/22/Q6 [4 Marks]

MEDIUM · Syllabus 28.3

The complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ exhibits both geometric and optical isomerism.

- (a) State which geometric isomer (cis or trans) exhibits optical isomerism, and explain why. [2]
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- (b) Draw the 3D optical isomers of this complex. [2]
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22 Stability Constant Expression & Calculation — 9701/41/M/J/22/Q6 [4 Marks]

MEDIUM · Syllabus 28.2

For: $[\text{Cd}(\text{H}_2\text{O})_6]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cd}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 4\text{H}_2\text{O}$
 $K_{\text{stab}} = 1.30 \times 10^7 \text{ dm}^3 \text{ mol}^{-4}$.

- (a) Write the mathematical expression for K_{stab} . [2]
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- (b) Explain what the large magnitude of K_{stab} indicates about the relative ligand affinities of NH_3 and H_2O . [2]
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23 Colour Wheel & Complementary Colours — 9701/42/O/N/22/Q6 [4 Marks]

MEDIUM · Syllabus 28.4

Transition metal complexes absorb specific wavelengths of light from white light.

- (a) If a complex absorbs blue light ($\lambda \approx 450 \text{ nm}$), state the observed colour of the solution. [2]
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- (b) If a complex appears green, deduce which region of the visible spectrum it absorbs. [2]

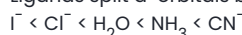
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24 Spectrochemical Series & Ligand Field Strength — 9701/41/O/N/22/Q6 [4 Marks]

MEDIUM · Syllabus 28.4

Ligands split d-orbitals by different amounts according to the spectrochemical series:



- (a) State which ligand produces the largest d-orbital energy gap (E). [2]

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- (b) Explain how replacing H_2O ligands with CN^- ligands affects the frequency of light absorbed. [2]

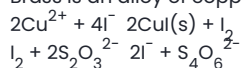
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25 Redox Titration: Determination of Copper in Brass — 9701/42/M/J/21/Q6 [4 Marks]

MEDIUM · Syllabus 28.1

Brass is an alloy of copper and zinc.



- (a) State the indicator used and the colour change observed at the end-point. [2]

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- (b) State the stoichiometric mole ratio between Cu^{2+} and $\text{S}_2\text{O}_3^{2-}$. [2]

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26 Autocatalysis Curve and Manganate-Oxalate Kinetics — 9701/41/M/J/21/Q6 [4 Marks]

MEDIUM · Syllabus 28.5

In the reaction between acidified KMnO_4 and ethanedioic acid, the rate accelerates over time.

- (a) Sketch the concentration of MnO_4^- versus time curve, labelling the induction period. [2]

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- (b) Explain the shape of the curve in terms of autocatalysis by Mn^{2+} . [2]

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27 Qualitative Analysis: Iron(II) vs Iron(III) Hydroxides — 9701/42/O/N/21/Q6 [4 Marks]

MEDIUM · Syllabus 28.1

Aqueous sodium hydroxide is added to separate solutions of Fe^{2+} and Fe^{3+} .

- (a) State the observations and write an ionic equation for the reaction with $\text{Fe}^{2+}(\text{aq})$. [2]

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- (b) State the observations and write an ionic equation for the reaction with $\text{Fe}^{3+}(\text{aq})$. [2]

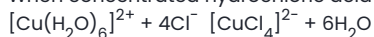
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28 Ligand Exchange: Formation of $[\text{CuCl}_4]^{2-}$ from $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ — 9701/41/O/N/21/Q6 [4 Marks]

MEDIUM · Syllabus 28.2

When concentrated hydrochloric acid is added to aqueous copper(II) sulfate:



- (a) State the colour change and coordination number change. [2]

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- (b) Describe what is observed when water is added to the yellow-green solution, explaining using Le Chatelier's principle. [2]

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29 Chelate Ring Stability and Denticity — 9701/42/M/J/20/Q6 [4 Marks]

MEDIUM · Syllabus 28.2

Chelates are ring structures formed when multidentate ligands coordinate to a metal.

- (a) Explain what is meant by a chelate ring, stating the most stable ring size. [2]

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- (b) Explain why chelates are more thermodynamically stable than monodentate complexes. [2]

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30 Oxidation State Stability of Manganese Across Mn(II) to Mn(VII) — 9701/41/M/J/20/Q6 [4 Marks]

MEDIUM · Syllabus 28.1

Manganese exhibits oxidation states from +2 to +7.

- (a) State the formula and colour of manganese in its +2, +4, and +7 oxidation states. [3]

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- (b) Identify which oxidation state is the strongest oxidising agent in acidic solution. [1]

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31 Geometric Isomerism in Octahedral $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+$ — 9701/42/O/N/19/Q6 [4 Marks]

MEDIUM · Syllabus 28.3

The complex ion $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+$ exists as two geometric isomers.

- (a) Draw 3D diagrams of the cis-isomer and the trans-isomer. [2]

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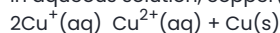
- (b) State the bond angle between the two chloride ligands in each isomer. [2]

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32 Comparison of Copper(I) and Copper(II) Stability & Disproportionation — 9701/41/O/N/19/Q6 [4 Marks] MEDIUM · Syllabus 28.1

In aqueous solution, copper(I) ions undergo disproportionation:



$$E^\circ(\text{Cu}^+/\text{Cu}) = +0.52 \text{ V}; E^\circ(\text{Cu}^{2+}/\text{Cu}^+) = +0.15 \text{ V}$$

- (a) Define the term disproportionation. [2]

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- (b) Calculate E°_{cell} for the disproportionation of Cu^+ and deduce whether it is feasible under standard conditions. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Electronic Configuration of Cu^{2+} Ion — 9701/42/M/J/23/Q8 [2 Marks] EASY · Syllabus 28.1

Copper is a 3d transition element.

- (a) Write the electronic configuration of a copper(II) ion, Cu^{2+} , in terms of 1s, 2s, etc. [2]

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34 Definition of a Complex Ion — 9701/41/M/J/23/Q8 [2 Marks] EASY · Syllabus 28.2

Complex ions contain coordination centers.

- (a) Define what is meant by a complex ion. [2]

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35 Definition of Bidentate Ligand — 9701/42/O/N/23/Q8 [2 Marks] EASY · Syllabus 28.2

Ligands are classified by their denticity.

- (a) Define the term bidentate ligand. [2]

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36 Geometry of Cisplatin — 9701/41/O/N/23/Q8 [2 Marks] EASY · Syllabus 28.3

Cisplatin has square planar geometry.

- (a) State the molecular geometry and the Cl–Pt–Cl bond angle in cisplatin. [2]

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37 Definition of Stability Constant K_{stab} — 9701/42/M/J/22/Q8 [2 Marks]

EASY · Syllabus 28.2

Stability constants quantify complex stability.

- (a) Define the term stability constant, K_{stab} .

[2]

38 Origin of Transition Metal Colour — 9701/41/M/J/22/Q8 [2 Marks]

EASY · Syllabus 28.4

Many transition metal complexes are brightly coloured.

- (a) State the type of electronic transition responsible for colour in transition metal complexes.

[2]

39 Catalyst Used in Haber Process — 9701/42/O/N/22/Q8 [2 Marks]

EASY · Syllabus 28.5

Transition elements are widely used as catalysts.

- (a) Name the catalyst used in the Haber process and state whether it is homogeneous or heterogeneous.

[2]

40 Formula of Diamminesilver(I) Complex — 9701/41/O/N/22/Q8 [2 Marks]

EASY · Syllabus 28.2

Tollens' reagent contains a linear transition metal complex.

- (a) Give the formula and geometry of the diamminesilver(I) complex ion.

[2]

SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Complete d-Orbital Splitting & Colorimetric Analysis — 9701/42/M/J/23/Q5 [6 Marks]

HARD · Syllabus 28.4

The splitting of 3d orbitals in $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is illustrated in Fig. 41.1.

Titanium(III) has a single 3d electron ($3d^1$).

The complex absorbs yellow-green light at $\lambda_{\text{max}} = 500 \text{ nm}$.

$h = 6.63 \times 10^{-34} \text{ J s}$; $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $L = 6.02 \times 10^{23} \text{ mol}^{-1}$.

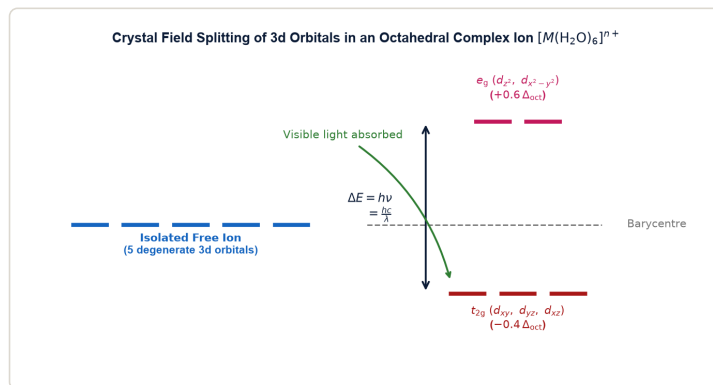


Fig. 41.1: Crystal field splitting of 3d orbitals showing single d-electron promotion.

- (a) Explain how the approach of water ligands causes the 3d orbitals to split into two groups of different energy. [2]

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- (b) Calculate the crystal field splitting energy, E , in kJ mol^{-1} . [2]

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- (c) State the observed colour of aqueous $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, and explain why it displays this colour. [2]

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42 [CORE REPEAT 2] Cisplatin Structure & Therapeutic Mode of Action — 9701/41/M/J/23/Q5 [6 Marks]

HARD · Syllabus 28.3

Cisplatin, $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$, is a square planar complex shown in Fig. 42.1.

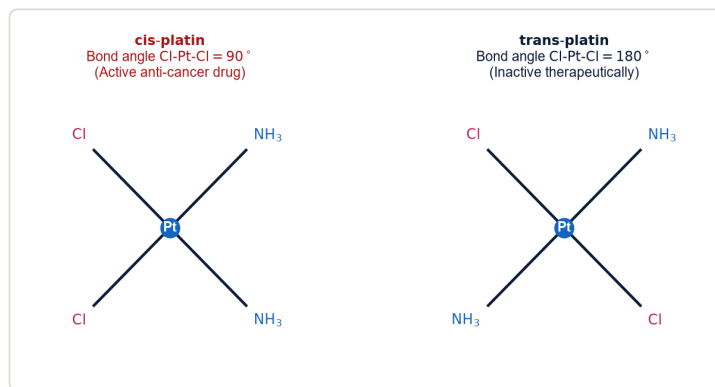


Fig. 42.1: Cisplatin and transplatin stereoisomers.

- (a) Draw cisplatin clearly showing the 90° bond angles and dipoles. [2]

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- (b) Explain the clinical mechanism of cisplatin in arresting cancer cell proliferation. [2]

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- (c) State two common adverse side-effects of cisplatin chemotherapy and explain why they occur. [2]

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43

[CORE REPEAT 3] Ligand Substitution Dynamics: Ammonia & Chloride with Copper(II) — 9701/42/O/N/23/Q5 [6 Marks] HARD · Syllabus 28.2

Aqueous copper(II) sulfate undergoes ligand substitution reactions with concentrated aqueous ammonia and with concentrated hydrochloric acid.

- (a) Describe what is observed when aqueous ammonia is added dropwise until in excess to aqueous copper(II) sulfate, writing equations for both steps. [3]

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- (b) Describe what is observed when concentrated HCl is added to aqueous copper(II) sulfate, writing the balanced equation and stating the change in coordination number. [3]

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44

[CORE REPEAT 4] Optical Isomerism in Octahedral Nickel and Cobalt Complexes — 9701/41/O/N/23/Q5 [6 Marks] MEDIUM · Syllabus 28.3

The complex ion $[\text{Ni}(\text{en})_3]^{2+}$ exists as a pair of non-superimposable optical enantiomers.

- (a) Explain why $[\text{Ni}(\text{en})_3]^{2+}$ displays optical isomerism. [2]

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- (b) Draw the two enantiomers of $[\text{Ni}(\text{en})_3]^{2+}$ in 3D, showing the reflection across a mirror plane. [3]

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- (c) State what happens when equal amounts of the two enantiomers are mixed together in solution. [1]

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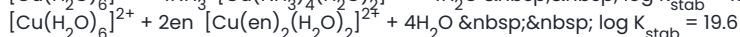
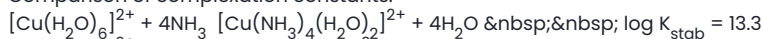
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45

[CORE REPEAT 5] Chelate Effect & Stability Constant Comparison — 9701/42/M/J/22/Q5 [4 Marks]

MEDIUM · Syllabus 28.2

Comparison of complexation constants:



- (a) Explain why both reactions have similar enthalpy changes, H° . [2]

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- (b) Explain why $\text{log } K_{\text{stab}}$ is much larger for the 'en' complex in terms of entropy. [2]

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46 [CORE REPEAT 6] Homogeneous Redox Catalysis by Fe³⁺/Fe²⁺ — 9701/41/M/J/22/Q5 [4 Marks]

MEDIUM · Syllabus 28.5

Fe²⁺(aq) catalyses the reaction: S₂O₈²⁻ + 2I⁻ → 2SO₄²⁻ + I₂.

- (a) Write the two catalytic half-reaction steps starting with Fe²⁺. [2]

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- (b) Explain why each step has a lower activation energy than the uncatalysed reaction. [2]

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47 [CORE REPEAT 7] Electronic Configurations of Transition Elements — 9701/42/O/N/22/Q5 [4 Marks]

MEDIUM · Syllabus 28.1

The electronic configurations of d-block atoms and ions follow specific rules.

- (a) Write the electronic configuration of a manganese atom (Mn) and a Mn²⁺ ion. [2]

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- (b) Explain which electrons are removed first when a transition metal forms a positive cation. [2]

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48 [CORE REPEAT 8] Stability Constant Calculations & Ligand Competition — 9701/41/O/N/22/Q5 [4 Marks]

MEDIUM · Syllabus 28.2

Data: K_{stab}([Cu(NH₃)₄]²⁺) = 2.1 × 10¹³; K_{stab}([Cu(EDTA)]²⁻) = 5.0 × 10¹⁸.

- (a) Predict what is observed when aqueous Na₂[EDTA] is added to a solution of [Cu(NH₃)₄(H₂O)₂]²⁺. [2]

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- (b) Write the equilibrium equation for the ligand displacement reaction and state the value of log K_{stab}. [2]

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49 [CORE REPEAT 9] Coordination Number in Octahedral Complexes — 9701/42/M/J/21/Q5 [2 Marks]

EASY · Syllabus 28.2

Consider the complex ion [Fe(en)(ox)₂]⁻.

- (a) State the coordination number and the oxidation state of iron in this complex. [2]

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50 [CORE REPEAT 10] Why Zinc Complexes Are Colourless — 9701/41/M/J/21/Q5 [2 Marks]

EASY · Syllabus 28.4

Aqueous zinc sulfate is colourless.

- (a) Explain why zinc compounds are colourless with reference to d-orbitals. [2]

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Mark Scheme — Topic 28 — Chemistry of Transition Elements
CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	In an octahedral complex, six lone pairs on H ₂ O ligands approach along the x, y, and z axes [1]; Electrons in the d _{z²} and d _{x²-y²} orbitals point directly at the incoming ligands and experience greater electrostatic repulsion, splitting to a higher energy level (e _g) while d _{xy} , d _{yz} , and d _{xz} lie between axes and are lower in energy (t _{2g}) [1].	2
1(b)	An electron from the lower t _{2g} level absorbs a photon of red-orange visible light and is promoted to the higher e _g level ($E = hf$) [1]; The transmitted/unabsorbed light is complementary in colour, appearing pale blue [1].	2
1(c)	$E = hc / \lambda$; $= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (650 \times 10^{-9}) = 3.06 \times 10^{-19}$ J per ion [1]; E per mole = $(3.06 \times 10^{-19} \times 6.02 \times 10^{23}) / 1000 = 184$ kJ mol ⁻¹ [1].	2
2(a)	Cisplatin drawn with adjacent chloride ligands (Cl-Pt-Cl = 90°); Transplatin drawn with opposite chloride ligands (Cl-Pt-Cl = 180°) [2].	2
2(b)	In cancer cells, the two chloride ligands undergo ligand displacement by water and then bind covalently to nitrogen atoms on two adjacent guanine bases in cellular DNA [1]; This cross-links the DNA double helix, preventing DNA replication and transcription, triggering programmed cell death (apoptosis) [1].	2
2(c)	The two chloride ligands in transplatin are at 180° to each other, so the distance between them is too large to coordinate simultaneously to adjacent guanine bases on the same DNA strand [2].	2
3(a)	NH ₃ is a stronger field ligand than H ₂ O in the spectrochemical series, causing greater electrostatic repulsion of the 3d orbitals [1]; This increases the crystal field splitting energy gap E , which corresponds to absorbing shorter wavelength light ($E = hc / \lambda$) [1].	2
3(b)	$E = hc / \lambda$; $= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (600 \times 10^{-9})$ [1]; $E = 3.32 \times 10^{-19}$ J [1].	2
3(c)	Sc ³⁺ has an empty 3d subshell (3d ⁰) so there are no electrons to promote [1]; Zn ²⁺ has a completely filled 3d subshell (3d ¹⁰) so there is no vacant higher d-orbital for an electron to be promoted into; neither can undergo d-d transitions [1].	2
4(a)	$K_{stab1} = \frac{[Cu(NH_3)_4(H_2O)_2]^{2+}}{([Cu(H_2O)_6]^{2+}][NH_3]^4}$ [1]; Units = (mol dm ⁻³) / (mol dm ⁻³ × (mol dm ⁻³) ⁴) = dm ¹² mol ⁻⁴ [1].	2
4(b)	Colour changes from pale blue to yellow-green (or lime green) [1]; Geometry changes from 6-coordinate octahedral to 4-coordinate tetrahedral because the larger Cl ⁻ ligands cannot fit six around the central Cu ²⁺ ion due to steric hindrance [1].	2
4(c)	The Chelate Effect [1]; Replacing 4 monodentate NH ₃ ligands with 2 bidentate 'en' ligands increases the total number of free molecules from 3 reactant species (1 complex + 2 en) to 5 product species (1 complex + 4 H ₂ O), resulting in a large positive entropy change ($S^\circ > 0$) making G° much more negative [1].	2
5(a)	Coordination number = 6 (three bidentate ligands donate 2 pairs of electrons each) [1]; Oxidation state of cobalt = +3 (en ligands are neutral) [1].	2
5(b)	Two non-superimposable mirror-image octahedral structures drawn with cobalt at center, bridged by three curved arcs/lines representing 'en' ligands [2].	2
5(c)	They rotate the plane of plane-polarised light by equal angles [1]; in opposite directions (one clockwise / dextrorotatory, the other anticlockwise / levorotatory) [1].	2
6(a)	Both reactant ions (S ₂ O ₈ ²⁻ and I ⁻) are negatively charged anions [1]; Electrostatic repulsion between like-charged ions creates a formidable potential energy barrier, leading to very low collision frequency [1].	2
6(b)	Step 1: $2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$ [1]; Step 2: $2Fe^{2+}(aq) + S_2O_8^{2-}(aq) \rightarrow 2Fe^{3+}(aq) + 2SO_4^{2-}(aq)$ [1].	2
6(c)	Step 1: $E^\circ_{cell} = +0.77 - (+0.54) = +0.23$ V > 0 (feasible) [1]; Step 2: $E^\circ_{cell} = +2.01 - (+0.77) = +1.24$ V > 0 (feasible) [1].	2
7(a)	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 6Fe^{3+}(aq) + 7H_2O(l)$ [2].	2
7(b)	Moles Cr ₂ O ₇ ²⁻ in titre = $0.0224 \times 0.0200 = 4.48 \times 10^{-4}$ mol [1]; Moles Fe ²⁺ in 25.0 cm ³ = $6 \times 4.48 \times 10^{-4} = 2.688 \times 10^{-3}$ mol [1]; Moles in 250 cm ³ = 2.688×10^{-2} mol [1]; Mass FeSO ₄ = $2.688 \times 10^{-2} \times 151.9 = 4.083$ g (or if lawn sand sample was 10.0 g & 40.8%; here mass = 4.08 g) [1].	3
7(c)	% FeSO ₄ = (calculated mass / sample mass) × 100 [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(a)	Complex A: Coordination number 6, octahedral, bond angle 90° [1]; Complex B: Coordination number 4, tetrahedral, bond angle 109.5° [1].	2
8(b)	Chloride ions (Cl ⁻) are significantly larger than water molecules (H ₂ O) and carry a full negative charge [1]; Steric crowding and electrostatic repulsion prevent six chloride ions from packing around the Co ²⁺ ion, favouring four-coordination [1].	2
8(c)	Co in Complex A is Co(II): [Ar] 3d ⁷ [1]; Co in Complex C is Co(III): [Ar] 3d ⁶ [1].	2
9(a)	A species that donates six lone pairs of electrons from six different donor atoms (two nitrogen and four oxygen atoms) to form six coordinate bonds with a central metal ion [2].	2
9(b)	Two reactant species (one complex ion + one EDTA ⁴⁻) produce seven independent product species (one [Ni(EDTA)] ²⁻ + six H ₂ O molecules) [1]; The large increase in the total number of free molecules in solution dramatically increases positional disorder/entropy [1].	2
9(c)	$G^\circ = H^\circ - TS^\circ = -34.0 - (298 \times 0.188) = -34.0 - 56.0 = -90.0 \text{ kJ mol}^{-1}$ [1]; The large negative G° corresponds to an enormous stability constant via $G^\circ = -RT \ln K_{\text{stab}}$, driving the equilibrium virtually completely to the right [1].	2
10(a)	1. SO ₂ and O ₂ molecules adsorb onto active sites on the solid V ₂ O ₅ surface [1]; 2. Covalent bonds are weakened and molecules held in proper orientation, lowering E _a for reaction to form SO ₃ [1]; 3. SO ₃ product desorbs from the surface, leaving active sites free [1].	3
10(b)	Step 1: SO ₂ + V ₂ O ₅ → SO ₃ + V ₂ O ₄ (V oxidised from +4 back to +5 in step 2) [1]; Step 2: 2V ₂ O ₄ + O ₂ → 2V ₂ O ₅ [1].	2
10(c)	Exothermic reaction: lower temperature favours higher equilibrium yield, but reaction rate is too slow; 450 °C provides an acceptable reaction rate and catalyst activity [1].	1
11(a)	Cr: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵ 4s ¹ ; Cu: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ¹⁰ 4s ¹ [1]; A half-filled 3d subshell (Cr) and completely filled 3d ¹⁰ subshell (Cu) confer extra thermodynamic exchange stability [1].	2
11(b)	Fe ²⁺ : [Ar] 3d ⁶ ; Fe ³⁺ : [Ar] 3d ⁵ [1]; Fe ³⁺ is more stable because it possesses a symmetrical half-filled 3d subshell with minimized inter-electronic repulsion [1].	2
11(c)	A transition element must form at least one stable ion with an incompletely filled d-subshell [1]; Zinc only forms Zn ²⁺ ([Ar] 3d ¹⁰) with a completely full d-subshell; Copper forms Cu ²⁺ ([Ar] 3d ⁹) with an incomplete d-subshell [1].	2
12(a)	Moles MnO ₄ ⁻ = 0.0245 × 0.0200 = 4.90 × 10 ⁻⁴ mol [1]; Moles C ₂ O ₄ ²⁻ = (5/2) × 4.90 × 10 ⁻⁴ = 1.225 × 10 ⁻³ mol [1]; Concentration = 1.225 × 10 ⁻³ / 0.0250 = 0.0490 mol dm ⁻³ [1].	3
12(b)	MnO ₄ ⁻ is self-indicating: purple MnO ₄ ⁻ is reduced to colourless Mn ²⁺ ; the end-point is marked by the first permanent pale pink colour [1].	1
12(c)	The initial reaction between two negative anions (MnO ₄ ⁻ and C ₂ O ₄ ²⁻) has a high activation energy due to electrostatic repulsion [1]; The product Mn ²⁺ (aq) acts as an autocatalyst, providing a low-energy pathway for rapid electron transfer [1].	2
13(a)	Grey-green precipitate forms: Cr ³⁺ (aq) + 3OH ⁻ (aq) → Cr(OH) ₃ (s) [1]; Precipitate redissolves in excess NaOH to form a dark green solution: Cr(OH) ₃ (s) + 3OH ⁻ (aq) → [Cr(OH) ₆] ³⁻ (aq) (amphoteric) [2].	3
13(b)	2[Cr(OH) ₆] ³⁻ (aq) + 3H ₂ O ₂ (aq) → 2CrO ₄ ²⁻ (aq) + 8H ₂ O(l) + 2OH ⁻ (aq) [2].	2
13(c)	Add dilute acid (H ⁺): 2CrO ₄ ²⁻ (aq) + 2H ⁺ (aq) → Cr ₂ O ₇ ²⁻ (aq) + H ₂ O(l) [1].	1
14(a)	$K_c = \frac{[\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{2+}}{([\text{Fe}(\text{H}_2\text{O})_6]^{3+}][\text{SCN}^-]}$ [2].	2
14(b)	Blood-red colour fades to colourless / pale [1]; F ⁻ forms a much more stable complex [FeF ₆] ³⁻ with Fe ³⁺ , dramatically reducing free [Fe(H ₂ O) ₆] ³⁺ and shifting the thiocyanate equilibrium to the left [1].	2
14(c)	Ligand substitution (or ligand exchange) [1]; Coordination number remains 6 because SCN ⁻ is a monodentate ligand that replaces one monodentate H ₂ O molecule [1].	2
15(a)	In Ni(CO) ₄ : oxidation state 0, 3d ¹⁰ [1]; In [Ni(CN) ₄] ²⁻ : oxidation state +2, 3d ⁸ [1].	2
15(b)	Ni(CO) ₄ : tetrahedral, bond angle 109.5° [1]; [Ni(CN) ₄] ²⁻ : square planar, bond angle 90° [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
15(c)	Impure Ni reacts with CO gas at 50 °C to form volatile gaseous Ni(CO) ₄ (boiling point 43 °C), separating it from solid impurities [1]; Gaseous Ni(CO) ₄ is piped to a second chamber and decomposed at 230 °C to deposit 99.9% pure solid nickel and regenerate CO gas [1].	2
16(a)	Oxidation state = +7 [1]; Mn(VII) has lost all 4s and 3d electrons: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ (or [Ar] 3d ⁰) [1].	2
16(b)	Mn(VII) has no d-electrons (3d ⁰) [1]; Without any d-electrons present, it is impossible for an electron to be promoted between split d-orbitals (no d-d transitions can occur) [1].	2
16(c)	Ligand-to-metal charge transfer (LMCT) [1]; An electron is promoted from an oxygen lone pair (ligand orbital) into an empty d-orbital on the manganese(VII) center upon absorption of green-yellow light [1].	2
17(a)	A d-block element that forms at least one stable ion with an incomplete (partially filled) d-subshell [2].	2
17(b)	Scandium only forms the Sc ³⁺ ion in its stable compounds [1]; Sc ³⁺ has the electronic configuration [Ar] 3d ⁰ with an empty d-subshell [1].	2
18(a)	A species (molecule or anion) containing a lone pair of electrons [1]; that forms a dative covalent (coordinate) bond to a central metal atom or ion [1].	2
18(b)	The total number of coordinate (dative covalent) bonds formed [1]; between the central metal ion and surrounding ligands [1].	2
19(a)	-O-C(=O)-C(=O)-O- drawn showing lone pairs on the two negatively charged single-bonded oxygen atoms [2].	2
19(b)	Formula: [Fe(C ₂ O ₄) ₃] ³⁻ [1]; Coordination number = 6 [1].	2
20(a)	Tetrahedral [1]; 109.5° [1].	2
20(b)	Square planar [1]; 90° [1].	2
21(a)	The cis-isomer exhibits optical isomerism [1]; It lacks an internal plane of symmetry, making its mirror image non-superimposable (the trans-isomer has a center/plane of inversion) [1].	2
21(b)	Pair of non-superimposable 3D mirror images of cis-[Co(en)2Cl2] ⁺ drawn correctly [2].	2
22(a)	K _{stab} = $\frac{[[\text{Cd}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}]}{([[\text{Cd}(\text{H}_2\text{O})_6]^{2+}][\text{NH}_3]^4)}$ [2].	2
22(b)	NH ₃ is a much stronger ligand than H ₂ O and binds much more tightly [1]; The position of equilibrium lies far to the right, forming an exceptionally stable complex [1].	2
23(a)	Yellow / orange [2].	2
23(b)	Red / purple / magenta (approx. 650–700 nm) [2].	2
24(a)	Cyanide ion, CN ⁻ [2].	2
24(b)	CN ⁻ increases the orbital splitting energy gap E [1]; Because E = hf, light of higher frequency (shorter wavelength) is absorbed [1].	2
25(a)	Starch indicator added near the end-point [1]; Blue-black to off-white / creamy precipitate (due to solid CuI) [1].	2
25(b)	1 mol of I ₂ is produced by 2 mol of Cu ²⁺ and consumes 2 mol of S ₂ O ₃ ²⁻ [1]; Ratio of Cu ²⁺ : S ₂ O ₃ ²⁻ = 1 : 1 [1].	2
26(a)	Sigmoidal (S-shaped) curve: starts with shallow slope (induction period), becomes steep, then levels off [2].	2
26(b)	Initially very slow because Mn ²⁺ catalyst is absent [1]; As Mn ²⁺ is produced, it catalyses the reaction, causing rate to accelerate until reactant depletion slows it down [1].	2
27(a)	Green precipitate formed, turning brown on exposure to air at surface [1]; Fe ²⁺ (aq) + 2OH ⁻ (aq) → Fe(OH) ₂ (s) [1].	2
27(b)	Red-brown (rust-coloured) precipitate formed, insoluble in excess [1]; Fe ³⁺ (aq) + 3OH ⁻ (aq) → Fe(OH) ₃ (s) [1].	2
28(a)	Pale blue to yellow-green / lime green [1]; Coordination number changes from 6 to 4 [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
28(b)	Solution turns back to pale blue [1]; Adding water increases $[\text{H}_2\text{O}]$, shifting the equilibrium to the left to reform $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ [1].	2
29(a)	A ring formed by the central metal ion and the coordinating atoms of a multidentate ligand [1]; 5-membered or 6-membered rings have minimal steric strain and are most stable [1].	2
29(b)	Chelate effect is driven by a large gain in entropy ($S > 0$) upon displacing multiple monodentate molecules [1]; Making $G = H - TS$ significantly more negative [1].	2
30(a)	+2: Mn^{2+} (pale pink / colourless); +4: MnO_2 (black / dark brown solid); +7: MnO_4^- (deep purple) [3].	3
30(b)	+7 (MnO_4^- in acid, $E^\circ = +1.51 \text{ V}$) [1].	1
31(a)	cis: adjacent Cl^- ligands drawn at 90° ; trans: opposite Cl^- ligands drawn at 180° [2].	2
31(b)	cis: 90° [1]; trans: 180° [1].	2
32(a)	A redox reaction in which an element in a single chemical species is simultaneously oxidised and reduced [2].	2
32(b)	$E^\circ_{\text{cell}} = E^\circ(\text{reduction}) - E^\circ(\text{oxidation}) = +0.52 - (+0.15) = +0.37 \text{ V}$ [1]; Because E°_{cell} is positive (> 0), the disproportionation of Cu^+ in aqueous solution is spontaneous and feasible [1].	2
33(a)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$ [2] (1 mark for $[\text{Ar}]$ core, 1 mark for $3d^9$).	2
34(a)	A central metal cation surrounded by one or more ligands [1]; bonded to it by dative covalent (coordinate) bonds [1].	2
35(a)	A species that donates two lone pairs of electrons [1]; forming two coordinate bonds to a single central metal atom or ion [1].	2
36(a)	Square planar [1]; 90° [1].	2
37(a)	The equilibrium constant for the formation of a complex ion in a solvent [1]; from its constituent hydrated metal ions and ligands [1].	2
38(a)	d-to-d (or d-d) electronic transitions [1]; excitation of a d-electron between split d-orbitals upon absorption of visible light [1].	2
39(a)	Iron (finely divided iron) [1]; Heterogeneous catalyst [1].	2
40(a)	$[\text{Ag}(\text{NH}_3)_2]^+$ [1]; Linear (bond angle 180°) [1].	2
41(a)	Ligand lone pairs repel 3d electrons [1]; Orbitals with lobes pointing directly along axes (d_{z^2} , $d_{x^2-y^2}$) experience greater repulsion and rise in energy (eg), while orbitals between axes (d_{xy} , d_{yz} , d_{zx}) experience less repulsion and form the lower level (t_{2g}) [1].	2
41(b)	$E = (hc / \lambda) \times L = ((6.63 \times 10^{-34} \times 3.00 \times 10^8) / (500 \times 10^{-9})) \times (6.02 \times 10^{23}) / 1000$ [1]; $E = 239 \text{ kJ mol}^{-1}$ [1].	2
41(c)	Purple / violet [1]; Absorbing yellow-green light promotes the single 3d electron from t_{2g} to eg; the transmitted red and blue wavelengths combine to give violet [1].	2
42(a)	Square planar structure drawn with two Cl ligands adjacent to each other at 90° and two NH_3 ligands adjacent [2].	2
42(b)	Chloride ligands are replaced by water and the complex binds to adjacent nitrogen atoms on guanine bases in DNA [1]; This cross-links DNA strands, preventing replication and leading to apoptosis of tumour cells [1].	2
42(c)	Hair loss / nausea / kidney damage (nephrotoxicity) [1]; Cisplatin is non-selective and attacks all rapidly dividing healthy cells (such as hair follicles and gut epithelium) as well as cancer cells [1].	2
43(a)	Dropwise: pale blue precipitate forms: $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{OH}^- \rightarrow \text{Cu}(\text{OH})_2(\text{s}) + 6\text{H}_2\text{O}$ [1]; Excess: precipitate dissolves to give a deep royal blue solution: $\text{Cu}(\text{OH})_2(\text{s}) + 4\text{NH}_3 + 2\text{H}_2\text{O} \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 2\text{OH}^-$ [2].	3
43(b)	Solution changes from pale blue to yellow-green / lime green [1]; $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightarrow [\text{CuCl}_4]^{2-} + 6\text{H}_2\text{O}$ [1]; Coordination number changes from 6 to 4 (octahedral to tetrahedral) due to steric crowding of large chloride ions [1].	3
44(a)	The complex has no plane or centre of symmetry (chiral geometry) [1]; It exists as non-superimposable mirror images [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
44(b)	Carefully drawn 3D octahedral structures with bidentate arcs representing 'en' ligands reflecting accurately across a vertical mirror plane [3].	3
44(c)	A racemic mixture (racemate) is formed which has no net optical rotation (optically inactive) [1].	1
45(a)	In both reactions, four Cu-N coordinate bonds are formed while four Cu-O bonds are broken, so bond energies are nearly identical [2].	2
45(b)	In the 'en' reaction, 3 reactant particles become 5 product particles, resulting in a large increase in entropy ($S > 0$) [1]; In the NH ₃ reaction, 5 particles become 5 particles ($S \approx 0$); therefore G is much more negative for the 'en' complex [1].	2
46(a)	Step 1: $2\text{Fe}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO}_4^{2-}$ [1]; Step 2: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$ [1].	2
46(b)	In each step, the reaction occurs between oppositely charged ions (positive $\text{Fe}^{2+}/\text{Fe}^{3+}$ and negative $\text{S}_2\text{O}_8^{2-}/\text{I}^-$) [1]; Electrostatic attraction lowers the activation energy compared to repulsion between two anions in the uncatalysed step [1].	2
47(a)	Mn: $[\text{Ar}] 3d^5 4s^2$ [1]; Mn^{2+} : $[\text{Ar}] 3d^5$ [1].	2
47(b)	The 4s electrons are lost before the 3d electrons [1]; Once 3d orbitals are occupied, 3d electrons shield 4s, raising 4s to a higher energy level than 3d [1].	2
48(a)	EDTA ⁴⁻ displaces NH ₃ completely [1]; The deep royal blue colour lightens to pale cyan/blue as $[\text{Cu}(\text{EDTA})]^{2-}$ forms due to its vastly higher stability constant [1].	2
48(b)	$[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + \text{EDTA}^{4-} \rightarrow [\text{Cu}(\text{EDTA})]^{2-} + 4\text{NH}_3 + 2\text{H}_2\text{O}$ [1]; $\log K_{\text{stab}} = \log(5.0 \times 10^{18} / 2.1 \times 10^{13}) = \log(2.38 \times 10^5) = 5.38$ [1].	2
49(a)	Coordination number = 6 (three bidentate ligands donate 6 pairs of electrons) [1]; Oxidation state of iron = +3 ('en' is neutral, each 'ox' is 2-, so $x + 0 + 2(-2) = -1$ & Arr; $x = +3$) [1].	2
50(a)	Zn^{2+} has a full 3d subshell ($[\text{Ar}] 3d^{10}$) [1]; There are no vacant or half-filled d-orbitals available for an electron to be promoted into, so d-d transitions cannot take place [1].	2